MILITARY SPECIFICATION SHEET

CONNECTORS, PRINTED CIRCUIT SUBASSEMBLY AND ACCESSORIES: RECEPTACLE, DECADE INCREMENT 10 THRU 110 CONTACT POSITIONS, FOR PRINTED WIRING BOARDS (.075 SPACING)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the connectors described herein shall consist of this specification and the latest issue of MIL-C-55302.

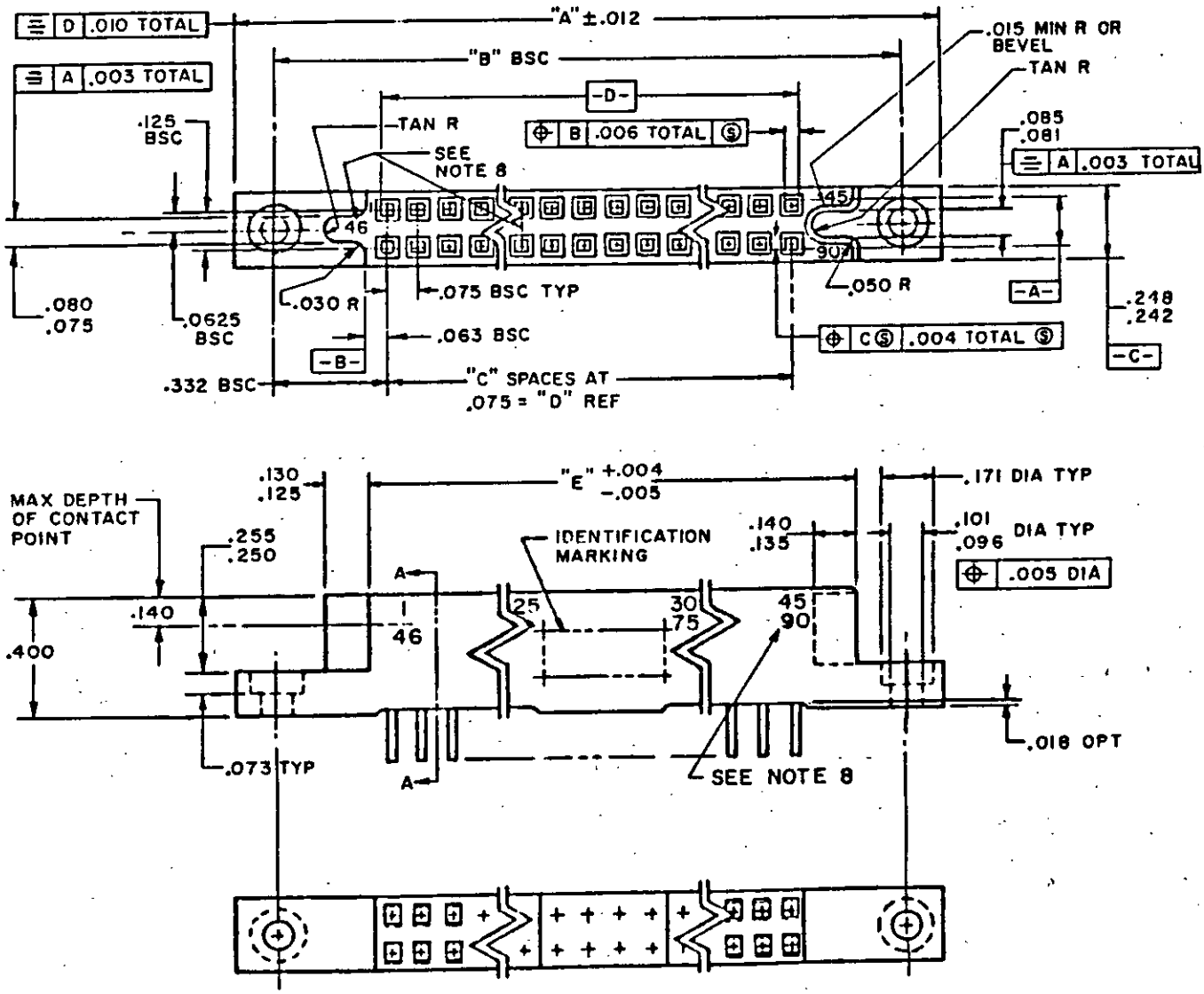


FIGURE 1. Connectors, receptacle (.075 spacing).

(E) denotes changes

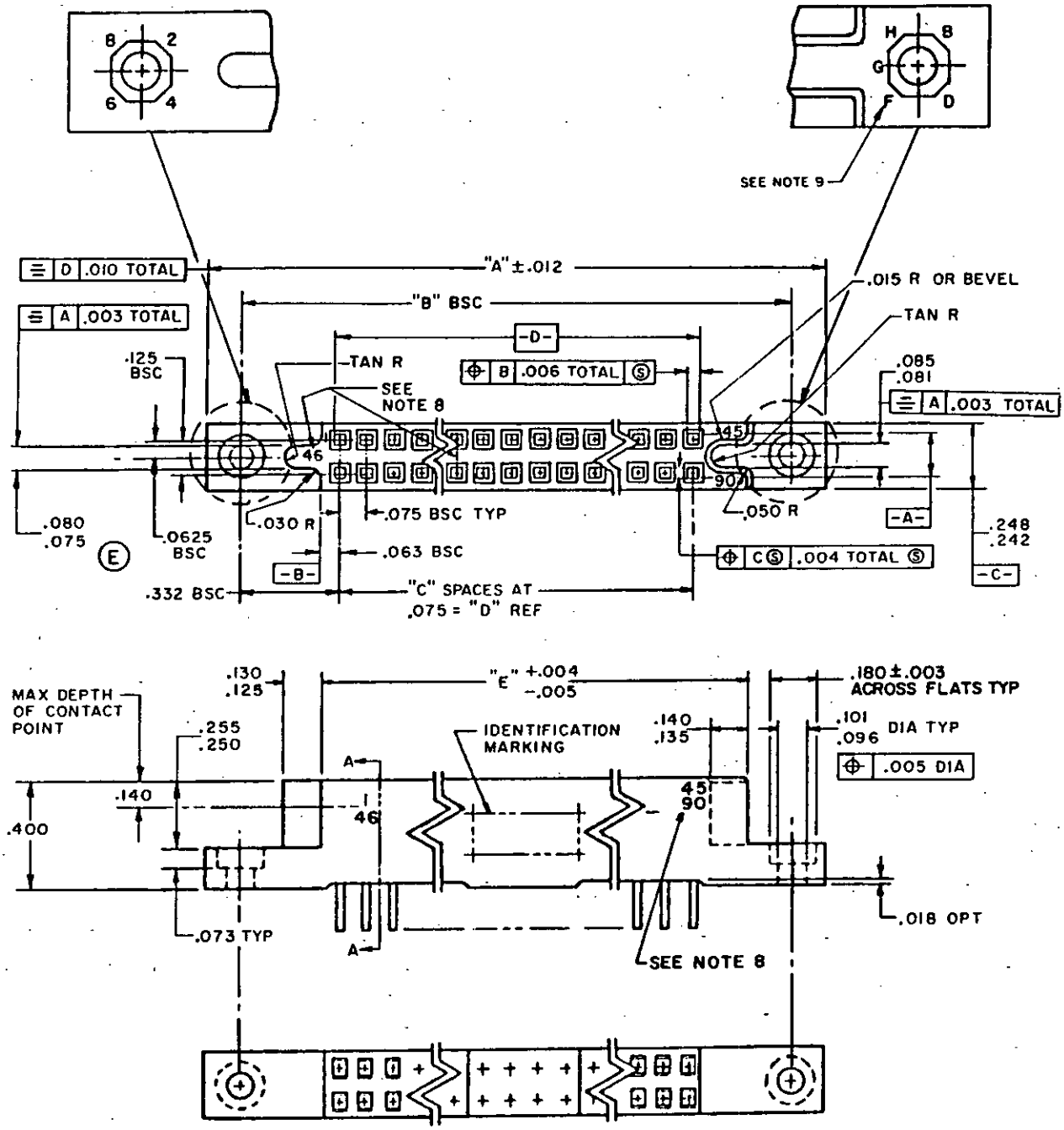
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1/ Dash number	Number of contacts	Type	A	B	C	D	E	Contact identification numbers
01 02 03 04	10	I II III IV	1.255 (31.88)	.965 (24.51)	4	.300 (7.62)	.565 (14.35)	1 - 5 6 - 10
05 06 07 08	20	I II III IV	1.630 (41.40)	1.340 (34.04)	9	.675 (17.15)	.940 (23.88)	1 - 10 11 - 20
09 10 11 12	30	I II III IV	2.005 (50.93)	1.715 (43.56)	14	1.050 (26.67)	1.315 (33.40)	1 - 15 16 - 30
13 14 15 16	40	I II III IV	2.380 (60.45)	2.090 (53.09)	19	1.425 (36.20)	1.690 (42.93)	1 - 20 21 - 40
17 18 19 20	50	I II III IV	2.755 (69.98)	2.465 (62.61)	24	1.800 (45.72)	2.065 (52.45)	1 - 25 26 - 50
21 22 23 24	60	I II III IV	3.130 (79.50)	2.840 (72.14)	29	2.175 (55.25)	2.440 (61.98)	1 - 30 31 - 60
25 26 27 28	70	I II III IV	3.505 (89.03)	3.215 (81.66)	34	2.550 (64.77)	2.815 (71.50)	1 - 35 36 - 70
29 30 31 32	80	I II III IV	3.880 (98.55)	3.590 (91.19)	39	2.925 (74.30)	3.190 (81.03)	1 - 40 41 - 80
33 34 35 36	90	I II III IV	4.255 (108.08)	3.965 (100.71)	44	3.300 (83.82)	3.565 (90.55)	1 - 45 46 - 90
37 38 39 40	100	I II III IV	4.630 (117.60)	4.340 (110.24)	49	3.675 (93.35)	3.940 (100.08)	1 - 50 51 - 100
41 42 43 44	110	I II III IV	5.005 (127.13)	4.715 (119.76)	54	4.050 (102.87)	4.315 (109.60)	1 - 55 56 - 110

1/ Dash numbers 01 through 44 are for style 1 (internal keying) connectors.

FIGURE 1. Connectors, receptacle, (.075 spacing) - Continued.

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STYLE 2 (EXTERNAL KEYING)

FIGURE 1. Connectors, receptacle,(.075 spacing) - Continued.

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1/ Dash number	Number of contacts	Type	A	B	C	D	E	Contact identification numbers
45 46 47 48	10	I II III IV	1.255 (31.88)	.965 (24.51)	4	.300 (7.62)	.565 (14.35)	1 - 5 6 - 10
49 50 51 52	20	I II III IV	1.630 (41.40)	1.340 (34.04)	9	.675 (17.15)	.940 (23.88)	1 - 10 11 - 20
53 54 55 56	30	I II III IV	2.005 (50.93)	1.715 (43.56)	14	1.050 (26.67)	1.315 (33.40)	1 - 15 16 - 30
57 58 59 60	40	I II III IV	2.380 (60.45)	2.090 (53.09)	19	1.425 (36.20)	1.690 (42.93)	1 - 20 21 - 40
61 62 63 64	50	I II III IV	2.755 (69.98)	2.465 (62.61)	24	1.800 (45.72)	2.065 (52.45)	1 - 25 26 - 50
65 66 67 68	60	I II III IV	3.130 (79.50)	2.840 (72.14)	29	2.175 (55.25)	2.440 (61.98)	1 - 30 31 - 60
69 70 71 72	70	I II III IV	3.505 (89.03)	3.215 (81.66)	34	2.550 (64.77)	2.815 (71.50)	1 - 35 36 - 70
73 74 75 76	80	I II III IV	3.880 (98.55)	3.590 (91.19)	39	2.925 (74.30)	3.190 (81.03)	1 - 40 41 - 80
77 78 79 80	90	I II III IV	4.255 (108.08)	3.965 (100.71)	44	3.300 (83.82)	3.565 (90.55)	1 - 45 46 - 90
81 82 83 84	100	I II III IV	4.630 (117.60)	4.340 (110.24)	49	3.675 (93.35)	3.940 (100.08)	1 - 50 51 - 100
85 86 87 88	110	I II III IV	5.005 (127.13)	4.715 (119.76)	54	4.050 (102.87)	4.315 (109.60)	1 - 55 56 - 110

1/ Dash numbers 45 through 88 are for style 2 (external keying) connectors.

FIGURE 1. Connectors, receptacle,(.075 spacing) - Continued.

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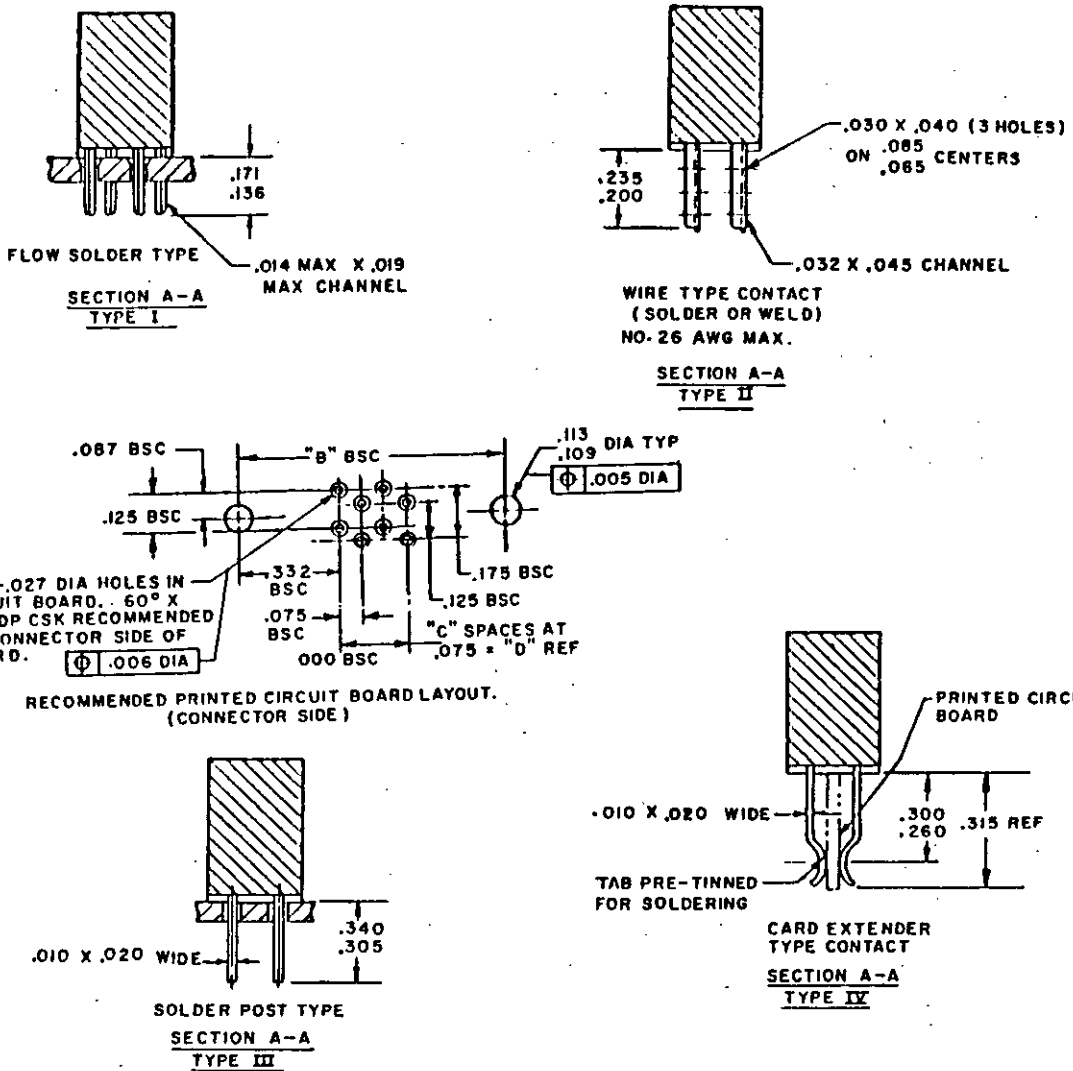


FIGURE 1. Connectors, receptacle,(.075 spacing) - Continued.

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Inches	mm	Inches	mm
.003	.08	.083	2.11
.004	.10	.085	2.16
.005	.13	.087	2.21
.006	.15	.096	2.44
.010	.25	.101	2.57
.012	.30	.109	2.77
.014	.36	.113	2.87
.015	.38	.125	3.18
.018	.46	.130	3.30
.019	.48	.135	3.43
.020	.51	.136	3.45
.022	.56	.140	3.56
.025	.64	.171	4.34
.027	.69	.175	4.44
.028	.71	.180	4.57
.030	.76	.200	5.08
.031	.79	.235	5.97
.032	.81	.242	6.15
.040	1.02	.248	6.30
.045	1.14	.250	6.35
.050	1.27	.255	6.48
.0625	1.588	.260	6.60
.063	1.60	.300	7.62
.065	1.65	.305	7.75
.067	1.70	.315	8.00
.073	1.85	.330	8.38
.075	1.90	.332	8.43
.080	2.03	.340	8.64
.081	2.06	.400	10.16

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.005 (.13 mm) on decimals and ±2° on angles.
4. These connectors mate with connectors specified in MIL-C-55302/23.
5. Use printed circuit boards with the type III receptacle.
6. Minimum pad length on printed circuit board.
7. Termination layout on .025 (.64 mm) modular grid.
8. Numbers indicating, every 5 cavities stamped on side (except 10 position). Numbers indicating end cavities and lines indicating every 5 cavities stamped on mating face.
9. Location indicators embossed on surface.

FIGURE 1. Connectors, receptacle,(.075 spacing) - Continued.

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REQUIREMENTS:

Design and construction:

Dimensions and configuration: See figure 1.

Material:
(E) Plating: Engagement area: Gold per MIL-C-55302 for .100 length, over nickel all over per MIL-C-55302.

Termination area: Tin lead over nickel per MIL-C-55302 for the following lengths:

.165 minimum length for type I.
.200 minimum length for type II.
.300 minimum length for type III.
.140 minimum length for type IV.

Contact identification: See figure 1.

Contact rating: 3.0 amperes maximum per contact, 2.25 amperes continuous per contact at 75°F.

Keying: See MIL-C-55302/31.

Oversized pin exclusion (sockets only): .0465 diameter pin.

Contact separation force: The connector shall meet the requirements of MIL-C-55302/29.

Wire size: Number 26 AWG maximum with type II, solder eyelet contact for wire termination.

Resistance to test probe damage: The connector shall meet the requirements of MIL-C-55302/30.

Mating and unmating: The maximum mating force in pounds shall be the number of contacts multiplied by .25 and the withdrawal force in pounds shall be a minimum of .025 times the number of contacts and shall not exceed the measured insertion force.

Contact resistance: The average contact resistance of all contacts measured shall not exceed .015 ohm, and no individual contact pair shall have a resistance exceeding .020 ohm.

Contact retention: 3 pounds minimum.

Dielectric withstanding voltage:
Sea level: 900 volts rms.
High altitude: 200 volts rms.

Replacement contacts: See MIL-C-55302/32.

Contact removal tool: MIL-I-81969/9-01.

Part number: M55302/24- (dash number from figure 1).

Patent number 3,404,367. The Government has a royalty free license under this patent for the benefit of manufacturers of the item either for the Government or for use in equipment to be delivered to the Government.

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Army - CR
Navy - EC
Air Force - 17

Review activities:
Army - CR, AR, MI
Navy - SH, AS, OS
Air Force - 11, 99
DLA - ES

User activities:
Army - ME, AT, AV
Navy - MC
Air Force - 19

Preparing activity:
Army - CR

Agent:
DLA - ES

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